

Rishabh Ravi

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🏠 University of Illinois, Urbana-Champaign

🌐 [Rishabh Ravi](#)

🐙 <http://github.com/borlaugg>

🌐 <https://borlaugg.github.io/>

Education

2025 – 30 📖 **University of Illinois, Urbana-Champaign, USA**
PhD in Electrical and Computer Engineering

2020 – 25 📖 **Indian Institute of Technology Bombay, India**
Bachelors and Masters of Technology in Electrical Engineering
Specializing in Electronic Systems (ES)
with Minor in Computer Science and Engineering

Publications

- 📖 SPARTON: Secure Dynamic Partition Scheme for Last-level Cache
Tejeshwar Thorawade, **Rishabh Ravi**, Keerthisagar Kokkiligadda, Nirmal Boran and Virendra Singh
Published in International Conference on Security and Cryptography (SECRYPT) 2026
- 📖 SCAM: Secure Shared Cache Partitioning Scheme
Varun Venkitraman, **Rishabh Ravi**, Tejeshwar Torwade, Nirmal Boran, Virendra Singh
Published in International Conference on Security and Cryptography (SECRYPT) 2025
- 📖 Hardware vs. Software Implementation of Warp-Level Features in Vortex RISC-V GPU
Huanzhi Pu, **Rishabh Ravi**, Shinnung Jeong, Udit Subramanya, Euijun Chung, Jisheng Zhao, Chihyo Ahn, Hyesoon Kim
Published in Open Source Solutions for Massively Parallel Integrated Circuits at DATE 2025

Professional Internships

- 2023 📖 **Modem Firmware Engineer** *Guide: Mr. Dileep Lingamallu | Qualcomm*
 - Revived a **GPRS-specific** modem firmware and virtual platform for applications in **smartwatches**
 - Created test case scenarios, emulating L1 commands, to assess and validate functionality by maintaining signal-to-noise ratio within defined **21 dB** thresholds, affirming the accuracy of decoded data
- 2022 📖 **Parallel Computing & Profiling** *Guide: Mrs. Tanmaya Karmakar | Nvidia*
 - Worked in a team to parallelize **ANUGA**, an open-source hydrodynamic modeling software, by profiling and running time analysis of the program on **Nvidia Nsight Systems**
 - Identified hot spots of the program that required parallelization to increase performance

Academic Achievements

- 2025 📖 **Ranked 1** in the ES specialization of Electrical Engineering based on academic performance
- 2024 📖 Received the **Undergraduate Research Award** for the work on cache side channel attacks
- 2023 📖 Designed a 2D Mapping System that ranked among the **top 3** projects out of 70+ projects.
- 2020 📖 Achieved All India Rank **878** in **JEE Advanced 2020** out of 150,000 candidates.
- 2020 📖 Secured **99.82 percentile** in **JEE Mains 2020** out of 1,100,000 candidates.

Mentorship

- 2023 - 24  **DAMP Mentor** *Student Mentorship Programme*
- Appointed as a mentor from a pool of 80+ applicants based on interviews and peer reviews. Mentored six sophomores on a one-to-one basis in their academic and co-curricular pursuits.
- 2023  **English Language Improvement Training TA** *Student Mentorship Programme*
- Assigned as an English Language Improvement Training one-on-one TA. Provided resources and held sessions that helped the students become more confident while speaking in public.
- 2022  **RnD Head** *AUV - IITB*
- Mentored a batch of recruits to AUV-IITB as the RnD head and helped train them. Provided resources and transferred knowledge regarding the functionality of the AUV.

Teaching

-  Teaching assistant at IIT Bombay for the following courses.

Year	Course	Instructor
2025	EE 309 Microprocessors	Prof. Rajababu V
2024	EE 709 Testing and Verification of VLSI Circuits (Head TA)	Prof. Madhav Desai
	EE 779 Advanced Topics in Signal Processing	Prof. Satish Mulleti
2023	MA 106 Linear Algebra (Head TA)	Prof. Jugal Verma
	MA 111 Integral Calculus	Prof. Preeti Raman
2022	MA 205 Complex Analysis	Prof. Saikat Mazumdar
	MA 207 Partial Differential Equations	Prof. Harsha Hutridurga